Lecture Notes

Lecture 2

Span, Linear Independence, Basis, Dimension

Last lecture a vector space was a set equipped with addition and scalar multiplication satisfying the vector-space axioms; a subset of an ambient vector space was a subspace exactly when it was closed under linear combinations. Today we turn linear combinations into the two most useful concepts in the subject. A *basis* is a minimal list of vectors from which every other vector is built in exactly one way; it is the abstract version of a coordinate system. The number of vectors in a basis is the *dimension*, the right notion of "size" for a vector space.

The pay-off is concrete. Once we fix a basis, every abstract vector—a polynomial, a matrix, a quantum state—becomes an honest column of numbers, and every linear-algebra computation reduces to arithmetic you already know. In quantum mechanics, choosing a basis is exactly what a physicist means by choosing a *representation*. Here $|\psi\rangle$ is only an early label for the vector ψ; Dirac notation is introduced formally in Lecture 7. Finite or discrete bases give ordinary coordinate columns. Position and momentum wavefunctions are the later continuous/generalized analogue, not finite columns in ordinary Hilbert bases. By the end of today you will be able to find bases, compute dimensions, pass between a vector and its coordinates, and count dimensions of sums and intersections.

Learning objectives

By the end of this lecture you will be able to:

- Compute the span of a list and recognise it as the smallest containing subspace.
- Test a list for linear independence using the linear-dependence lemma.
- Find a basis and dimension, and pass between a vector and its coordinate column.
- Apply the dimension formula $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.

1 Span

Definition 2.1 (Span). The *span* of a list $v_1, \ldots, v_m$ in V is the set of all their linear combinations,

$$\operatorname{span}(v_1, \ldots, v_m) = \{a_1 v_1 + \cdots + a_m v_m : a_1, \ldots, a_m \in \mathbb{F}\}.$$

By convention the span of the empty list is $\{0\}$.

Proposition 2.2 (The span is the smallest containing subspace). $\operatorname{span}(v_1, \ldots, v_m)$ is a subspace of V, and it is contained in every subspace of V that contains $v_1, \ldots, v_m$.

Proof. It contains 0 (take all scalars 0), and a sum or scalar multiple of linear combinations of the v_k is again such a combination, so the subspace test (Lecture 1) is met. If a subspace U contains every v_k, then it contains all their linear combinations by closure, i.e. $U \supseteq \operatorname{span}(v_1, \ldots, v_m)$. ■

Definition 2.3 (Spanning list, finite-dimensional). If $\operatorname{span}(v_1, \ldots, v_m) = V$ we say

$v_1, \ldots, v_m$ *spans* V. A vector space is *finite-dimensional* if some finite list spans it, and *infinite-dimensional* otherwise.

Example 2.4 (Standard spanning lists). The list $e_1 = (1, 0, \ldots, 0), \ldots, e_n = (0, \ldots, 0, 1)$ spans $\mathbb{F}^n$, since $(x_1, \ldots, x_n) = x_1 e_1 + \cdots + x_n e_n$; so $\mathbb{F}^n$ is finite-dimensional. Likewise $1, t, \ldots, t^m$ spans $\mathcal{P}_m(\mathbb{F})$. By contrast $\mathcal{P}(\mathbb{F})$ is *infinite*-dimensional. An empty or all-zero finite list does not even span 1. Any other finite list has a maximal degree d among its nonzero members, so its combinations cannot produce t^{d+1}. ◁

Geometrically, the span of a single nonzero vector is a line through the origin, and the span of two non-parallel vectors in $\mathbb{R}^3$ is a plane through the origin (Figure 1): every linear combination stays inside it.

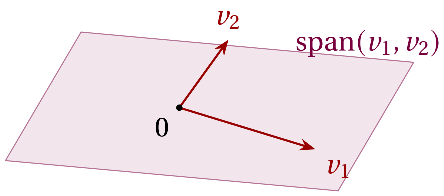

Figure 1: Two non-parallel vectors in $\mathbb{R}^3$ span a plane through the origin.

Example 2.5 (Is a vector in the span?). Is $(15, 16, 17)$ in $\operatorname{span}\big((1, 2, 3), (6, 5, 4)\big)$ in $\mathbb{R}^3$? We seek scalars a, b with $a(1, 2, 3) + b(6, 5, 4) = (15, 16, 17)$, i.e. the system

$$a + 6b = 15, \qquad 2a + 5b = 16, \qquad 3a + 4b = 17.$$

From the first two, $a = 15 - 6b$ and $2(15 - 6b) + 5b = 16$ give $b = 2$, $a = 3$; the third equation $3(3) + 4(2) = 17$ checks. Hence $(15, 16, 17) = 3(1, 2, 3) + 2(6, 5, 4)$ and the answer is *yes*. Deciding membership in a span is exactly solving a linear system—the Gauss–Jordan machinery from your prerequisites. ◁

2 Linear independence

A spanning list may carry redundancy: some vector may already be a combination of the others. The lists with *no* redundancy are the important ones.

Definition 2.6 (Linear independence). A list $v_1, \ldots, v_m$ is *linearly independent* if the only scalars $a_1, \ldots, a_m$ with

$$a_1 v_1 + \cdots + a_m v_m = 0$$

are $a_1 = \cdots = a_m = 0$. Otherwise it is *linearly dependent*: there exist scalars, not all zero, giving 0. (The empty list is independent.)

Remark 2.7. Independence is precisely the statement that each vector of the span has a *unique* expression as a linear combination: if $\sum a_k v_k = \sum b_k v_k$ then $\sum (a_k - b_k) v_k = 0$, so independence forces $a_k = b_k$. This is why bases will give unique coordinates.

Example 2.8 (Independent and dependent lists). The standard list $e_1, \ldots, e_n$ is independent in $\mathbb{F}^n$, as is $1, t, \ldots, t^m$ in $\mathcal{P}_m(\mathbb{F})$. A length-one list is independent iff its vector is nonzero; a length-two list is independent iff neither vector is a scalar multiple of the other. A dependent example in $\mathbb{F}^3$:

$$2(2, 3, 1) + 3(1, -1, 2) + (-1)(7, 3, 8) = (0, 0, 0).$$

And from analysis, the three functions $1,\ \cos^2 t,\ \sin^2 t$ are linearly *dependent* in $\mathbb{R}^{\mathbb{R}}$, because $1 - \cos^2 t - \sin^2 t = 0$ for all t. ◁

Example 2.9 (Testing independence by a determinant). For which c are the vectors $(1, 2, 3)$, $(0, 1, 4)$, $(2, 3, c)$ independent in $\mathbb{R}^3$? Three vectors in $\mathbb{R}^3$ are independent exactly when the matrix with these rows is invertible, i.e. has nonzero determinant. Expanding along the first column,

$$\det\begin{pmatrix}1 & 2 & 3\\ 0 & 1 & 4\\ 2 & 3 & c\end{pmatrix} = 1\,(c-12) - 0 + 2\,(8-3) = c - 12 + 10 = c - 2.$$

So the list is linearly independent precisely when $c \neq 2$, and dependent when $c = 2$. The determinant test (and, equivalently, row reduction) is the practical way to decide independence of vectors given by coordinates. ◁

The next lemma is the technical engine behind everything in this lecture: in a dependent list, some vector is redundant and can be thrown away.

> **Lemma 2.10 (Linear dependence lemma).** If $v_1, \ldots, v_m$ is linearly dependent, then there is an index j with $v_j \in \operatorname{span}(v_1, \ldots, v_{j-1})$, and removing v_j from the list leaves the span unchanged. (For $j = 1$ this reads $v_1 = 0$.)

Proof. Dependence gives scalars $a_1, \ldots, a_m$, not all zero, with $\sum a_i v_i = 0$. Let j be the largest index with $a_j \neq 0$. If $j = 1$ then $a_1 v_1 = 0$ with $a_1 \neq 0$, so $v_1 = 0 \in \{0\} = \operatorname{span}(\,)$. If $j > 1$ then

$$v_j = -\tfrac{1}{a_j}\big(a_1 v_1 + \cdots + a_{j-1} v_{j-1}\big) \in \operatorname{span}(v_1, \ldots, v_{j-1}).$$

For the span claim, any combination of $v_1, \ldots, v_m$ becomes a combination of the remaining vectors once we substitute this expression for v_j; hence deleting v_j does not shrink the span (and clearly does not enlarge it). ■

Example 2.11 (Finding the redundant vector). Take $(1, 2, 3), (6, 5, 4), (15, 16, 17), (8, 9, 7)$ in $\mathbb{R}^3$. We will soon see $\mathbb{R}^3$ is three-dimensional, so a list of four vectors must be dependent; the dependence lemma then says some vector is a combination of the earlier ones. Which is the first? The opening vector is nonzero, so $j = 1$ fails, and the second is not a scalar multiple of the first, so $j = 2$ fails. For $j = 3$ we solve $(15, 16, 17) = a(1, 2, 3) + b(6, 5, 4)$ and find $a = 3,\ b = 2$ (exactly Example 2.5). Thus $j = 3$ is the first index at which a vector lies in the span of its predecessors, and deleting $(15, 16, 17)$ leaves the span unchanged. ◁

3 The fundamental counting theorem

Everything that follows—that dimension is well defined, that bases behave as coordinate systems—rests on one inequality.

> **Theorem 2.12 (Independent ≤ spanning).** In a vector space V, the length of any linearly independent list is at most the length of any spanning list.

Proof. Let $u_1, \ldots, u_m$ be independent and $w_1, \ldots, w_n$ span V; we prove $m \leq n$. We run a process of m steps, each inserting one u and deleting one w, keeping a spanning list of length n throughout.

Step 1. The list $w_1, \ldots, w_n$ spans V, so adjoining u_1 in front gives the dependent list $u_1, w_1, \ldots, w_n$ (here u_1 lies in the span of the w's). By Lemma 2.10 some entry is in the span of the earlier ones; it is not u_1, since independence makes $u_1 \neq 0$ and $\operatorname{span}(\,) = \{0\}$. So it is some w_k, which we delete. The result is a spanning list of length n containing u_1.

Step k ($2 \le k \le m$). Suppose we hold a spanning list of length n containing $u_1, \ldots, u_{k-1}$. Insert u_k just after u_{k-1}: the new list is dependent (every added vector lies in the span of a spanning list). By Lemma 2.10 some entry is in the span of the earlier ones. It cannot be any of $u_1, \ldots, u_k$, for those are independent and so none lies in the span of the previous u's. Hence it is one of the remaining w's, which we delete, restoring a spanning list of length n that now contains $u_1, \ldots, u_k$.

If $m > n$, then after Step n we hold a spanning list of length n consisting of $u_1, \ldots, u_n$ *only* (all w's gone). Step $n+1$ inserts u_{n+1}, producing a dependent list of u's alone; Lemma 2.10 then puts some u in the span of earlier u's, contradicting independence. Therefore $m \le n$. ■

A useful by-product, proved by the same idea (keep choosing vectors outside the current span until Theorem 2.12 stops you), is that every subspace of a finite-dimensional space is itself finite-dimensional.

4 Bases and dimension

Definition 2.13 (Basis). A *basis* of V is a list that is linearly independent and spans V.

Theorem 2.14 (Bases give unique coordinates). A list $e_1, \ldots, e_n$ is a basis of V if and only if every $v \in V$ has a unique expression $v = a_1 e_1 + \cdots + a_n e_n$ with $a_k \in \mathbb{F}$.

Proof. ($\Rightarrow$) Spanning gives existence; independence gives uniqueness, by the remark in §2. ($\Leftarrow$) Existence of such expressions for every v says the list spans. Applying uniqueness to $0 = \sum a_k e_k$ (which also equals $\sum 0 \cdot e_k$) forces every $a_k = 0$, i.e. independence. So the list is a basis. ■

Example 2.15 (Standard bases). $e_1, \ldots, e_n$ is a basis of $\mathbb{F}^n$ (the *standard basis*); $1, t, \ldots, t^m$ is a basis of $\mathcal{P}_m(\mathbb{F})$; and the matrix units E_{ij} (a 1 in entry (i, j), zeros elsewhere) form a basis of $M_{m\times n}(\mathbb{F})$. ◁

A basis can always be produced, in two complementary ways: by trimming a spanning list down, or by growing an independent list up.

Theorem 2.16 (Reduction to a basis). Every spanning list of V can be reduced to a basis of V by deleting some of its vectors.

Proof. While the list is linearly dependent, Lemma 2.10 supplies a vector lying in the span of the earlier ones; delete it, which by the same lemma leaves the span equal to V. Each deletion shortens a finite list, so the process halts, and it can halt only when the list is independent—an independent spanning list, i.e. a basis. ■

Theorem 2.17 (Extension to a basis). In a finite-dimensional space, every linearly independent list extends to a basis.

Proof. Let $u_1, \ldots, u_m$ be independent and $w_1, \ldots, w_n$ a basis. Apply Theorem 2.16 to the spanning list $u_1, \ldots, u_m, w_1, \ldots, w_n$. No u_i is ever deleted: the vectors preceding it are $u_1, \ldots, u_{i-1}$, which are independent, so u_i is not in their span and is not a candidate for removal. Only w's are discarded, leaving a basis that still contains $u_1, \ldots, u_m$. ■

Corollary 2.18 (Existence of a basis). Every finite-dimensional vector space has a basis—reduce any finite spanning list.

Example 2.19 (Reducing a spanning list). The list $(1,0),(0,1),(1,1)$ spans $\mathbb{R}^2$ but is dependent, since $(1,1)=(1,0)+(0,1)$. That last vector lies in the span of the earlier two, so the reduction of Theorem 2.16 deletes it, leaving the basis $(1,0),(0,1)$. ◁

Example 2.20 (Extending an independent list). To extend $(1,1,0,0),(0,1,1,0)$ to a basis of $\mathbb{R}^4$, append the standard basis $e_1,\dots,e_4$ and drop redundant vectors as in Theorem 2.17. One outcome is

$$(1,1,0,0),\ (0,1,1,0),\ (1,0,0,0),\ (0,0,0,1),$$

which is linearly independent and spans $\mathbb{R}^4$, hence a basis. ◁

For dimension to make sense, all bases of a space must have the same length. Theorem 2.12 delivers exactly this.

Theorem 2.21 (Basis length is well defined). Any two bases of a finite-dimensional vector space have the same length.

Proof. Let $\mathcal{B}_1,\mathcal{B}_2$ be bases. Since $\mathcal{B}_1$ is independent and $\mathcal{B}_2$ spans, Theorem 2.12 gives $\operatorname{len}\mathcal{B}_1 \le \operatorname{len}\mathcal{B}_2$. Swapping their roles gives the reverse inequality, so the lengths are equal. ■

Definition 2.22 (Dimension). The *dimension* $\dim V$ of a finite-dimensional vector space is the length of any basis.

Example 2.23 (Dimensions). $\dim \mathbb{F}^n = n$; $\dim \mathcal{P}_m(\mathbb{F}) = m+1$; $\dim M_{m\times n}(\mathbb{F}) = mn$. And, as promised in Lecture 1, the field $\mathbb{C}$ has $\dim_{\mathbb{C}} \mathbb{C} = 1$ but $\dim_{\mathbb{R}} \mathbb{C} = 2$ (basis $1, i$): the dimension depends on the field of scalars, not just on the set. ◁

Two shortcuts make checking a basis far easier once the dimension is known; both follow from Theorem 2.12.

Proposition 2.24 (Right length is enough). Let $\dim V = n$. Then any linearly independent list of length n is a basis, and any spanning list of length n is a basis.

Proof. An independent list of length n extends to a basis by Theorem 2.17; but every basis has length n, so no vector is added and the list is already a basis. A spanning list of length n reduces to a basis by Theorem 2.16; again every basis has length n, so no vector is removed and the list is already a basis. ■

Proposition 2.25 (Subspaces have smaller dimension). If V is finite-dimensional and $U \subseteq V$ is a subspace, then $\dim U \le \dim V$, with equality if and only if $U = V$.

Proof. A basis of U is an independent list in V, so $\dim U \le \dim V$ by Theorem 2.12. If $\dim U = \dim V = n$, a basis of U is an independent list of length n in V, hence a basis of V by Proposition 2.24; thus $U = \operatorname{span}(\text{that list}) = V$. ■

Example 2.26 (A basis of a subspace by solving a system). Let $U = \{x \in \mathbb{R}^4 : x_1+x_2+x_3+x_4 = 0,\ x_1 - x_3 = 0\}$. Solving, $x_1 = x_3$ and $x_2 = -x_1 - x_3 - x_4 = -2x_1 - x_4$, leaving x_1 and x_4 free. With $(x_1, x_4) = (1, 0)$ and $(0, 1)$ we read off

$$U = \operatorname{span}\big((1, -2, 1, 0),\ (0, -1, 0, 1)\big).$$

These two vectors are independent (neither is a multiple of the other), so they form a basis and $\dim U = 2$. Finding a basis of a solution space is just back-substitution after row reduction. ◁

5 Coordinates relative to a basis

Theorem 2.14 lets us attach numbers to abstract vectors.

Definition 2.27 (Coordinate vector). Let $\mathcal{B} = (e_1, \dots, e_n)$ be an *ordered* basis of V. The *coordinate vector* of $v = a_1 e_1 + \cdots + a_n e_n$ is $[v]_{\mathcal{B}} = (a_1, \dots, a_n) \in \mathbb{F}^n$.

Proposition 2.28 (The coordinate map). The map $v \mapsto [v]_{\mathcal{B}}$ is a bijection $V \to \mathbb{F}^n$ that respects the operations: $[v + w]_{\mathcal{B}} = [v]_{\mathcal{B}} + [w]_{\mathcal{B}}$ and $[av]_{\mathcal{B}} = a\,[v]_{\mathcal{B}}$.

Proof. Bijectivity is Theorem 2.14 (existence and uniqueness of coordinates). Reading off coefficients is additive and homogeneous because the basis expansion is. ■

This proposition is the bridge between the abstract and the computational. Once a basis is chosen, *every* n-dimensional space over $\mathbb{F}$ looks like $\mathbb{F}^n$. We will name this phenomenon—*isomorphism*—in Lecture 4. The coordinates of v depend on the basis, as Figure 2 shows.

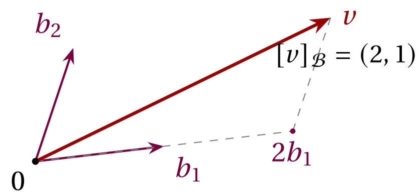

Figure 2: Coordinates relative to a (non-orthogonal) basis $\mathcal{B} = (b_1, b_2)$: reach v by going 2 steps along b_1 and 1 along b_2.

Example 2.29 (Coordinates of a polynomial). In $\mathcal{P}_2(\mathbb{R})$ take $p(t) = 3 + 2t - t^2$. In the standard basis $\mathcal{B} = (1, t, t^2)$ we read $[p]_{\mathcal{B}} = (3, 2, -1)$. In the shifted basis $C = \big(1,\ t - 1,\ (t - 1)^2\big)$, substitute $s = t - 1$ (so $t = s + 1$):

$$p = 3 + 2(s + 1) - (s + 1)^2 = 3 + 2s + 2 - (s^2 + 2s + 1) = 4 - s^2,$$

so $[p]_C = (4, 0, -1)$. Same polynomial, different coordinate columns. ◁

Example 2.30 (Coordinates by solving a system). Take the basis $\mathcal{B} = \big((1, 1), (1, -1)\big)$ of $\mathbb{R}^2$ and the vector $v = (4, 2)$. Writing $v = a(1, 1) + b(1, -1)$ gives $a + b = 4$ and $a - b = 2$, so $a = 3$, $b = 1$ and $[v]_{\mathcal{B}} = (3, 1)$—even though in the standard basis $[v] = (4, 2)$. Finding coordinates in a given basis is, once again, solving a linear system. ◁

Remark 2.31 (Bases as representations). As a preview of Dirac notation, $|v\rangle$ denotes the same vector as v. For a finite or discrete basis a basis expansion reads $|\psi\rangle = \sum_k a_k |e_k\rangle$, and the coordinate vector $(a_1, \dots, a_n)$ is the "wavefunction" of the state in that basis. In the continuous position or momentum representation, the analogue is a function and requires the Hilbert-space machinery of Lectures 13–14. Changing representation changes the coordinates while leaving the represented pure-state ray untouched. Linear algebra is the bookkeeping that makes such changes routine.

6 The dimension formula for sums

Recall from Lecture 1 that $U + W$ and $U \cap W$ are subspaces. Their dimensions are linked by a counting formula, analogous to inclusion–exclusion for sets.

Theorem 2.32 (Dimension of a sum). If U and W are subspaces of a finite-dimensional vector space, then

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

Proof. Choose a basis $v_1, \dots, v_p$ of $U \cap W$, so $\dim(U \cap W) = p$. Being independent in U, it extends to a basis $v_1, \dots, v_p, x_1, \dots, x_q$ of U (so $\dim U = p + q$), and to a basis $v_1, \dots, v_p, y_1, \dots, y_r$ of W (so $\dim W = p + r$). We claim

$$v_1, \dots, v_p,\ x_1, \dots, x_q,\ y_1, \dots, y_r \tag{2.1}$$

is a basis of $U + W$, which finishes the proof since then $\dim(U + W) = p + q + r = (p + q) + (p + r) - p$.

The list (2.1) clearly spans $U + W$. For independence, suppose

$$\sum_i \alpha_i v_i + \sum_j \beta_j x_j + \sum_k \gamma_k y_k = 0.$$

Put $z = \sum_i \alpha_i v_i + \sum_j \beta_j x_j$; then $z \in U$ and also $z = -\sum_k \gamma_k y_k \in W$, so $z \in U \cap W$. Hence z is a combination of the v_i alone; comparing with $z = \sum_i \alpha_i v_i + \sum_j \beta_j x_j$ and using independence of the basis of U forces every $\beta_j = 0$. The original relation becomes $\sum_i \alpha_i v_i + \sum_k \gamma_k y_k = 0$, and independence of the basis of W forces all $\alpha_i = \gamma_k = 0$. Thus (2.1) is independent. ∎

Corollary 2.33 (Direct sums add dimensions). $\dim(U \oplus W) = \dim U + \dim W$. More generally, $U + W$ is direct if and only if $\dim(U + W) = \dim U + \dim W$.

Proof. By Theorem 2.32, $\dim(U + W) = \dim U + \dim W$ exactly when $\dim(U \cap W) = 0$, i.e. $U \cap W = \{0\}$, which is the two-subspace criterion for directness from Lecture 1. ∎

Example 2.34 (Two planes must meet). If U, W are two-dimensional subspaces (planes through 0) of $\mathbb{R}^3$, then $\dim(U \cap W) = \dim U + \dim W - \dim(U + W) \geq 2 + 2 - 3 = 1$. So two planes through the origin in $\mathbb{R}^3$ always intersect in at least a line—they can never meet only at 0. ◁

Example 2.35 (Computing $\dim(U + W)$ and $\dim(U \cap W)$). In $\mathbb{R}^4$ let

$$U = \operatorname{span}\big((1, 1, 0, 0), (0, 0, 1, 1)\big), \qquad W = \operatorname{span}\big((1, 0, 1, 0), (0, 1, 0, 1)\big),$$

both of dimension 2. Row-reducing the four vectors together shows three of them are independent and the fourth is a combination of the others, so $\dim(U + W) = 3$. The formula then gives

$$\dim(U \cap W) = \dim U + \dim W - \dim(U + W) = 2 + 2 - 3 = 1.$$

Indeed $(1, 1, 1, 1) = (1, 1, 0, 0) + (0, 0, 1, 1) \in U$ and also $(1, 1, 1, 1) = (1, 0, 1, 0) + (0, 1, 0, 1) \in W$, so $U \cap W = \operatorname{span}\big((1, 1, 1, 1)\big)$, confirming the count. ◁

Summary of main concepts

The span of a list is the smallest subspace containing it. A list is *independent* when 0 has only the trivial representation, and a *basis* is an independent spanning list—equivalently, a list giving every vector unique coordinates. Every linearly independent list is no longer than any spanning list (Theorem 2.12); hence all bases share a length, the *dimension*. In an n-dimensional space, an independent or spanning list of length n is automatically a basis. Fixing an ordered basis turns each vector into a coordinate column in $\mathbb{F}^n$; in a finite/discrete model this is what a physicist calls choosing a representation. Continuous position and momentum representations instead produce wavefunctions and require the later Hilbert-space framework. Finally, $\dim(U+W) = \dim U + \dim W - \dim(U \cap W)$, with equality to $\dim U + \dim W$ exactly for direct sums.

- **Span**—all linear combinations of a list; the smallest subspace containing it.
- **Linear independence**—only the trivial combination yields 0.
- **Basis**—an independent spanning list; gives every vector unique coordinates.
- **Dimension**—the common length of all bases (independent $\leq$ spanning).
- **Coordinate vector**—the column $[v]_{\mathcal{B}} \in \mathbb{F}^n$ once a finite ordered basis is fixed; continuous representations are function-valued analogues.
- **Dimension formula**—$\dim(U+W) = \dim U + \dim W - \dim(U \cap W)$; additive exactly for direct sums.

Exercises for this lecture are in the companion sheet `exercises-2.pdf`.